

Restaurant Management System - Bug Fixing Guide

This document outlines the intentional bugs present in the system and hints for fixing them.

Overview

This Flask application is intentionally broken with various common bugs that students can identify and fix. The system implements a basic restaurant management system with menu management, order taking, and bill generation functionality.

Bug Categories

1. Route inconsistencies
2. Logic errors in calculations
3. Missing functionality
4. Incorrect data handling
5. Template rendering issues

Main Bugs to Fix

App Configuration Issues

1. **Missing Secret Key:** The application is missing a secret key, which is needed for session management.
 - Hint: Add `app.secret_key = 'restaurant_secret_key'` to enable session functionality.

Data Management Issues

1. **Missing Save Function:** The application doesn't save data back to the JSON files.
 - Hint: Uncomment and implement the `save_data()` function.
2. **No File Existence Check:** The `initialize_data` function doesn't properly check if files exist.
 - Hint: Fix the file handling in the `initialize_data` function.

Route Issues

1. **Typo in Route Path:** There's a typo in the add menu item route ('/meu/add' instead of '/menu/add').
 - Hint: Fix the route path in the decorator.

2. **Missing Delete Route:** The `delete_menu_item` route is completely missing, but is referenced in templates.
 - Hint: Uncomment and implement the `delete_menu_item` route.
3. **Incorrect Redirect:** The redirect after creating a new order uses string concatenation instead of `url_for`.
 - Hint: Replace with proper `url_for` call.

Data Processing Issues

1. **Price Not Converted to Float:** The item price is not converted to float when adding or editing menu items.
 - Hint: Use `float(request.form.get('price'))` instead of just getting the value.
2. **Inefficient Search Algorithm:** The `edit_menu_item` function uses an inefficient search algorithm.
 - Hint: Refactor the search to be more efficient.
3. **Incorrect Item Removal:** The `remove_item_from_order` function incorrectly removes items.
 - Hint: Use proper list indexing with `pop()` instead of `remove()`.
4. **Missing Total Calculation:** Order totals are not calculated or updated properly.
 - Hint: Implement proper total calculation when adding/removing items.
5. **Incorrect Tax Calculation:** The tax calculation in the bill view is incorrect.
 - Hint: Review and fix the calculation.

Form Validation Issues

1. **Missing Input Validation:** There's no validation for price being a number or table number being valid.
 - Hint: Add proper validation in the POST handling code.
2. **Category Validation:** The `add_menu_item` function doesn't validate the category.
 - Hint: Add validation to ensure the category exists in the menu structure.

Template Issues

1. **Typo in Category Value:** In `add_menu_item.html`, there's a typo in the "appetizers" category value.
 - Hint: Fix the value to match the key used in the menu dictionary.
2. **Missing Delete Route in Menu Template:** The menu template references a non-existent delete route.
 - Hint: Implement the missing route or update the template.
3. **Incorrect Parameter in View Order:** The `remove_item_from_order` link uses incorrect parameters.
 - Hint: Fix the parameter naming to match what the function expects.

Bonus Challenges

Once the basic bugs are fixed, consider these additional improvements:

1. Implement proper error handling throughout the application
2. Add session management for multiple users
3. Implement user roles (manager, waiter, etc.)
4. Add a search functionality for menu items and orders
5. Implement proper date/time formatting for orders
6. Add analytics (daily/weekly sales reports)
7. Implement item categories management

Testing Your Fixes

After fixing a bug, test the application by:

1. Running the Flask application (`python app.py`)
2. Navigate to the affected functionality in the web browser
3. Verify the feature now works as expected
4. Check that the data is properly saved and maintained between server restarts

Good luck fixing the bugs!